

DEEFAKE DETECTION USING ARTIFICIAL INTELLIGENCE

Spatio-Temporal Hybrid Deep Learning Framework (ResNeXt50 + LSTM)

K. D. Polytechnic, Patan | Department of Computer Engineering | Diploma Poster Competition 2024-25



1. SHOWCASE PORTAL (VERCEL)

<https://deepfakedetection-cyan.vercel.app/>

Live 24x7 Hosted on Vercel:

Judges can scan with any phone to view the interactive System Architecture, 8K Forensics, and Benchmark Analytics.



2. LIVE AI ENGINE (NGROK)

<https://petite-discover-precut.ngrok-free.dev/>

Live PyTorch Detection Engine:

Scan to upload video right in front of judges and see instant real-time REAL/FAKE prediction & 68 facial landmarks.

5-MINUTE PRESENTATION TIMELINE & POSTER STAGE CUES

Timing	Presentation Stage	What to Point on Poster	Key Target Message
0:00 - 1:00	Stage 1: Opening & Problem	Poster Title & Section 1	The Deepfake Crisis: Seeing is no longer believing!
1:00 - 2:15	Stage 2: ResNeXt50 + LSTM	Section 2 & Figure 1 (Split)	Why Spatio-Temporal beats single-frame CNNs.
2:15 - 3:30	Stage 3: 4-Step Pipeline	Section 4 & Figure 2 (LSTM)	20-frame sampling to 112x112 to Softmax verdict.
3:30 - 4:15	Stage 4: Live Demo & QR	Section 8 (Showcase QR)	Invite judges to scan on their smartphones.
4:15 - 5:00	Stage 5: Benchmarks & Ablation	Section 9, 10, 11 & Matrix	87% accuracy, +8.6% LSTM boost, SOTA comparison.

PRESENTER GOLDEN RULES FOR 1ST PRIZE:

- Smile & Confident Eye Contact:** Look at all judges with a warm smile. Never turn your back to the judges.
- Point Clearly with Hand/Pen:** When saying "Section 1", "Figure 1", "Figure 2", or "Showcase QR", point clearly to the poster.
- Natural GujEnglish Flow:** Speak comfortably in easy conversational Gujarati and pronounce English technical terms clearly.

STAGE 1: OPENING & THE PROBLEM (0:00 - 1:00)

[ACTION: Judges samaksha smile karo ane Namaste kari speech start karo]

"Respected Judges and Faculty Members, a very warm Good Morning to all of you!

Aaj na digital era ma ek famous proverb che je badlai gayu che: **Seeing is no longer believing!**

FaceSwap ane DeepFaceLab jeva modern AI generative tools etla realistic videos banave che ke naked human eye thi Real ane Fake vachhe farak parkhavo impossible bani gayo che.

Poster na **Section 1** ma tame joi shako cho: Deepfakes identity theft, financial fraud, ane fake news ma **900% annual surge** sathe failai rahya che, jyare human detection rate **54% thi pan ochho** che.

Aa national cyber threat ne solve karva mate ame develop karyu che: **Deepfake Detection Using Artificial Intelligence!**"

STAGE 2: CORE NOVELTY - RESNEXT50 + LSTM (1:00 - 2:15)

[ACTION: Poster na Section 2 ma Figure 1 (Split Face) taraf point karo]

"Judges, normal systems khali single image (photo) check kare che. Pan deepfake creators single photo ma blending perfect kari de che!

Pan video ma **Temporal Continuity (frame-to-frame flow)** preserve nathi kari shakta - jem ke eye blinking, head turning jitter, ane lighting flicker!

Aetle amari main innovation che: **Hybrid Spatio-Temporal Framework!**

- **1. ResNeXt-50 (Spatial Backbone):** Grouped Convolutions thi facial boundaries ane skin pore artifacts extract kare che.
- **2. LSTM (Temporal Network):** 20 continuous frames ni series ma micro-flickering ane optical flow drift pakde che.

Poster na **Figure 1** ma tame joi shako cho: Real side par biological skin coherent che, jyare Fake side par **Grad-CAM Heatmap** boundary artifacts ne clear highlight kare che!"

STAGE 3: END-TO-END SYSTEM PIPELINE (2:15 - 3:30)

[ACTION: Poster na Center Column ma Section 4 & 5 na Flowchart taraf point karo]

"Amari system nu execution pipeline clear 4 steps ma work kare che:

- **Step 1 (Upload):** User Django web portal par video upload kare che.
- **Step 2 (20-Frames Sampling):** OpenCV video mathi uniform **20 frames (1-second temporal horizon)** extract kare che.
- **Step 3 (Face Crop):** Dlib 68-landmarks localize kari ne face 112x112 pixels ma normalize kare che.
- **Step 4 (Hybrid Inference):** ResNeXt-50 2048-dim features banave che, Bi-LSTM sequence evaluate kare che, ane final result display kare che: **REAL ke FAKE + Confidence % + Heatmap!**"

STAGE 4: LIVE DEMO & SHOWCASE QR CODE (3:30 - 4:15)

[ACTION: Poster na Section 8 ma rahela Showcase QR Code taraf point karo]

"Judges, amaro project khali theoretical concept nathi, fully deployed working system che!

Poster na Section 8 ma **amari Showcase Website no Official QR Code** che, je 24/7 Vercel cloud par live hosted che. Tame potana mobile thi scan kari ne interactive architecture, 8K forensics heatmaps, ane test analytics live verify kari shako cho!

Client side par real-time **68-point facial landmark mesh** pan in-browser WebGL sathe render thay che!"

STAGE 5: BENCHMARKS, ABLATION STUDY & CLOSING (4:15 - 5:00)

[ACTION: Section 9, 10 na Tables ane Section 6 na 87% Accuracy Box taraf point karo]

"Ame model ne 23,186 videos na 3 world-class datasets par train karyu che: **DFDC (Meta AI), Celeb-DF v2, ane FaceForensics++**.

- **SOTA Comparison (Section 9):** MesoNet-4 (70.4%), ResNet-50 (78.4%), ane XceptionNet (83.2%) karta amaru Hybrid Model **87% Accuracy** achieve kare che!
- **Ablation Study (Section 10):** Single-frame baseline karta Bi-LSTM  **+8.6%**  che.
- **Defense (Section 11):** FaceSwap (89.2%) ane DeepFaceLab (86.8%) banne against high defense provide kare che.

Real-World Applications: Banking E-KYC video spoof protection, Newsroom media verification, ane Cyber crime forensics.

Thank you so much for your time and kind attention! We are now open for your valuable questions!"

STAGE LIVE DEMO ACTION CHECKLIST (IF JUDGES ASK TO RUN DEMO):

- **Video 1 (Real):** Upload TESTING VIDEO/1_REAL_Male_Interview.mp4 → Show green verdict: **REAL (High Confidence)**.
- **Video 2 (Fake):** Upload TESTING VIDEO/2_FAKE_Male_Deepfake.mp4 → Show red alert: **FAKE (Deepfake Detected)**.
- **Landmark Live Canvas:** Point out 68 neon cyan facial dots moving across frames on the browser screen in real time.

TOP 6 JUDGES Q&A CHEAT-SHEET (EASY 2-LINE WINNING ANSWERS)

Judges aa questions maathi j prashno poochhse. Aa concise javabo bold ane confident aapo:

Q01: ResNeXt-50 kem pasand karvu? Regular ResNet kem nahi?

Sir, ResNeXt-50 ma **Cardinality (32 parallel grouped convolution paths)** hoy che. Deepfake artifacts micro-level na hoy che, jene ResNeXt independent paths thi regular ResNet karta superior spatial precision thi pakde che.

Q02: LSTM nu shu kaam che aa project ma?

Sir, CNN khali static 2D photo samje che, pan video temporal media che! Deepfakes ma inter-frame eye blinking ane boundary jitter ma discontinuity hoy che. **LSTM 20 frames na sequence ma aa temporal inconsistency catch kare che (+8.6% gain).**

Q03: Video mathi 20 frames j kem lidhi? Badhi 300 frames kem nahi?

Sir, standard video ma 300 frames hoy che, jene process karta 15-20 seconds thai jaay. **20 frames thi inference khali 1.85 seconds ma thai jaay che ane 87% accuracy male che.** Aa latency ane accuracy no optimal sweet spot che.

Q04: XceptionNet ane EfficientNet ■■■■ ■■■■■■ ■■■■■■ ■■■ ■■■■■■ ■■■?

Sir, XceptionNet ane EfficientNet khali single-frame spatial features check kare che. Deepfake creators single frame ma blending perfect kari le che, pan 20 frames ni temporal continuity ma jitter aavi jaay che, je amaru LSTM pakdi paade che.

Q05: Input resolution 112x112 kem raakhvu? 224x224 kem nahi?

Sir, background remove karya pachi face crop mate 112x112 resolution purti che. **224x224 karta 112x112 ma GPU memory usage 75% ochhi thai jaay che,** jena lidhe amaru model fast ane lightweight bane che.

Q06: FaceSwap ane DeepFaceLab ma difference shu che ane tamari system banne ne detect kare che?

Sir, FaceSwap outer boundary artifacts chodi de che (Section 11 ■■■■ 89.2% detection), jyare DeepFaceLab latent feature blur kare che (86.8% detection). Amaru model banne generative methods ne high accuracy thi pakde che.